

Question 1*(12 marks)***Marks**

- (a) Factorize $2m^3 - 128$ 2
- (b) Simplify $\frac{1}{y-x} + \frac{x}{(x-y)^2}$ 3
- (c) Solve $\frac{x^2-4}{x} \geq 0$ 3
- (d) Solve $x-y=6$ and $x^2+y^2=18$.
Explain the geometric significance of your answer. 4

Question 2*(12 marks)***Start this Question on a new page.**

- (a) If $x-y+\sqrt{x+y}=\sqrt{6}$ find x and y . 4
- (b) If $x=\frac{3-\sqrt{3}}{3+\sqrt{3}}$ find $x-\frac{1}{x}$ and hence $x^2+\frac{1}{x^2}$. 4
- (c) (i) If $\phi(x)=(x-1)^2$ show that $\phi(P+1)=\phi(1-P)$ 4
(ii) If $f(x)=x^2+2$ and $F(x)=2x+3$, find $f(F(x))$ and $F(f(x))$.

Question 3*(12 marks)***Start this Question on a new page.**

- (a) Sketch $y=|x+2|$ and $y=|x-4|$ on the same axes
and then use your graph to solve $|x+2| \leq |x-4|$ 3
- (b) Shade the intersection of the regions represented by
 $x^2+y^2 > 1$ and $x^2+y^2 \leq 9$ 3
- (c) Shade the region $y < \sqrt{9-x^2}$ keeping in mind any
restrictions on the domain. 3
- (d) If $\cot \theta = \frac{3}{4}$ and $180^\circ < \theta < 360^\circ$ use exact values to show that
 $\frac{9 \sec \theta \tan \theta}{1 + \sin \theta} = -100$ 3

Question 4

(12 marks)

Start this Question on a new page.

- (a) Prove $\frac{1+\cos\theta}{1-\cos\theta} = (\operatorname{cosec}\theta + \cot\theta)^2$ 3
- (b) Eliminate θ from the pair of equations $x = 2 + 3\cos^2\theta$ 2
 $y = 2\sin\theta - 5$
- (c) Solve for $0^\circ \leq x \leq 360^\circ$ (i) $\sin\frac{x}{2} = \frac{1}{2}$ 3
- (ii) $2\cos^2\theta - 3\sin\theta - 3 = 0$ 4

Question 5

(12 marks)

Start this Question on a new page.

- (a) (i) Find the size of the largest angle of a triangle with sides 3cm, 4cm and 6cm. 3
- (ii) Hence find the area of the triangle. 2
- (b) From a point A, on the deck of a ship, the angle of elevation of an aircraft (P) is 34° and from an observation point B, 25m vertically above A, the angle of elevation is $32^\circ 30'$. Calculate, to the nearest 10m, the height of the aircraft above the deck. 7

Question 4*(12 marks)***Start this Question on a new page.****Marks**

(a) Prove $\frac{1 + \cos \theta}{1 - \cos \theta} = (\operatorname{cosec} \theta + \cot \theta)^2$ 3

(b) Eliminate θ from the pair of equations $x = 2 + 3 \cos^2 \theta$ 2
 $y = 2 \sin \theta - 5$

(c) Solve for $0^\circ \leq x \leq 360^\circ$ (i) $\sin \frac{x}{2} = \frac{1}{2}$ 3

(ii) $2 \cos^2 \theta - 3 \sin \theta - 3 = 0$ 4

Question 5*(12 marks)***Start this Question on a new page.**

(a) (i) Find the size of the largest angle of a triangle with sides 3cm, 4cm and 6cm. 3

(ii) Hence find the area of the triangle. 2

(b) From a point A, on the deck of a ship, the angle of elevation of an aircraft (P) is 34° and from an observation point B, 25m vertically above A, the angle of elevation is $32^\circ 30'$.

Calculate, to the nearest 10m, the height of the aircraft above the deck. 7

Question 6

(12 marks)

Start this Question on a new page.

- (a) Noting any restrictions on the domain of the function sketch

$$y = \frac{x^2 - 9}{x + 3}$$

2

- (b) Examine
- $y = \frac{1 + 2x^2}{1 - x^2}$
- noting

- any restrictions on the domain
- symmetry
- intercepts
- asymptotes both horizontal and vertical

and then draw the sketch.

4

- (c) For the curve
- $y = f(x) = \frac{1}{\sqrt{x-1}}$

- (i) By considering the natural domain, any intercepts and the y values as
- $x \rightarrow 1^+$
- and
- $x \rightarrow \infty$
- sketch the curve.

3

- (ii) Find the inverse function
- $f^{-1}(x)$
- and sketch this on the same axes.

3

Question 1

$$(a) 2m^3 - 128 = 2(m^3 - 64) \\ = 2(m-4)(m^2 + 4m + 16)$$

$$(b) \frac{1}{y-x} + \frac{x}{(x-y)^2}$$

$$= \frac{-1(x-y)}{(x-y)(x-y)} + \frac{x}{(x-y)^2}$$

$$= \frac{-x + y + x}{(x-y)^2}$$

$$= \frac{y}{(x-y)^2}$$

$$c) x^2 - 4 \geq 0 \quad x \neq 0$$

$$x(x-2)(x+2) \geq 0$$

$$x-2 \leq x \leq 0 \text{ and } x \geq 2$$



$$) x - y = 6 \quad \text{--- (1)}$$

$$x^2 + y^2 = 18 \quad \text{--- (2)}$$

arrange (1)

$$x = y + 6$$

$$(y+6)^2 + y^2 = 18$$

$$y^2 + 12y + 36 + y^2 = 18$$

$$y^2 + 12y + 18 = 0$$

$$y^2 + 6y + 9 = 0$$

$$(y+3)^2 = 0$$

$$y = -3$$

sub in (1)

$$x - 3 = 6$$

$$x = 9$$

$$\text{check in (2)} \quad 9^2 + (-3)^2 = 18 \checkmark$$

Solution: $x = 9, y = -3$

geometric significance.

There is one point of intersection between a line and a circle. The line is a tangent to the circle.

Question 2

$$b) x - y + \sqrt{x+y} = \sqrt{6}$$

$$(x - y) + \sqrt{x+y} = 0 + \sqrt{6}$$

$$\therefore x - y = 0 \quad \text{--- (1)}$$

$$x + y = 6 \quad \text{--- (2)}$$

Solve simultaneously.

$$[1 + 2] \quad 2x = 6 \\ \therefore x = 3$$

$$[\text{sub } x \text{ in (1)}] \quad 3 - y = 0 \\ y = 3$$

$$[\text{check in (2)}] \quad 3 + 3 = 6 \checkmark$$

$$\therefore x = 3 \text{ and } y = 3$$

$$c) i) \phi(x) = (x-1)^2$$

$$\phi(p+1) = (p+1-1)^2 \\ = p^2$$

$$\phi(1-p) = (1-p-1)^2 \\ = (-p)^2 \\ = p^2$$

$$\therefore \phi(p+1) = \phi(1-p)$$

$$iii) f(x) = x^2 + 2$$

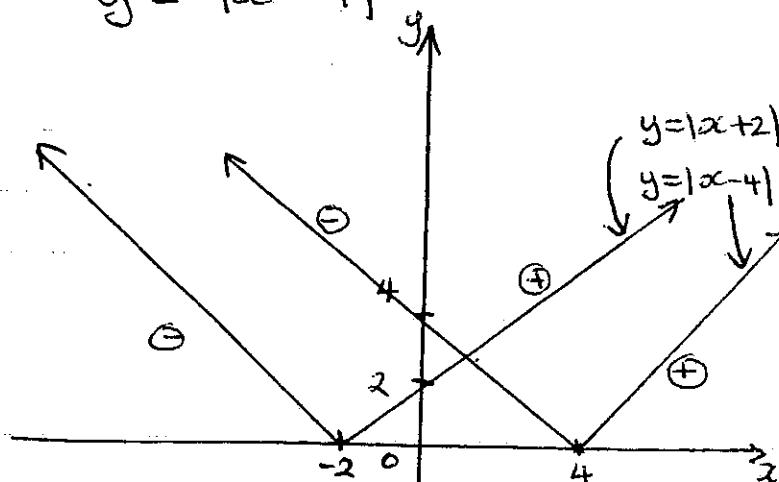
$$F(x) = 2x + 3$$

$$\text{find } f(F(x)) = (2x+3)^2 + 2 \\ = 4x^2 + 6x + 9 + 2 \\ = 4x^2 + 6x + 11$$

$$F(f(x)) = 2(x^2 + 2) + 3 \\ = 2x^2 + 4 + 3 \\ = 2x^2 + 7$$

Question 3

(a) $y = |x+2|$
 $y = |x-4|$



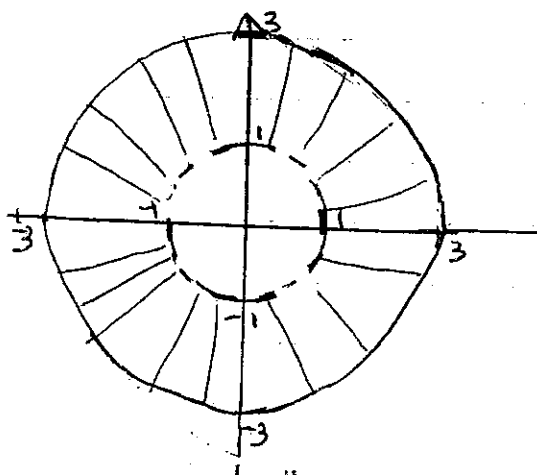
point of intersection

$$\begin{aligned} + (x+2) &= - (x-4) \\ x+2 &= -x+4 \\ 2x &= 2 \\ \therefore x &= 1 \end{aligned}$$

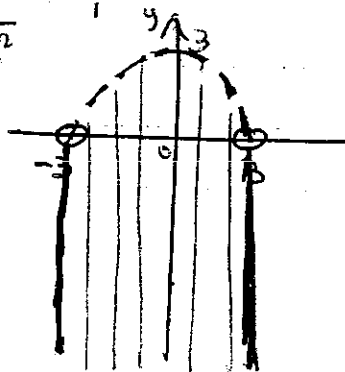
For $|x+2| \leq |x-4|$
 $\therefore x \leq 1$

(b) $x^2 + y^2 > 1$
 centre (0,0)
 radius 1

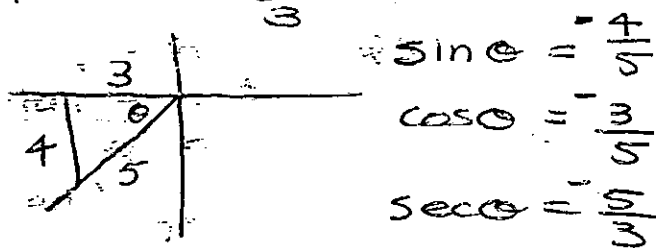
$x^2 + y^2 \leq 9$
 centre (0,0)
 radius 3



(c) $y < \sqrt{9-x^2}$



(d) $\cot \theta = \frac{3}{4}$ $+180^\circ \leq \theta \leq 360^\circ$
 $\therefore \tan \theta = \frac{4}{3}$



$\sin \theta = -\frac{4}{5}$

$\cos \theta = -\frac{3}{5}$

$\sec \theta = -\frac{5}{3}$

$$\begin{aligned} \text{LHS} &= \frac{9 \sec \theta \tan \theta}{1 + \sin \theta} \\ &= \frac{9 \times \frac{5}{3} \times \frac{4}{3}}{1 + \frac{-4}{5}} \end{aligned}$$

$$\begin{aligned} &= \frac{-20}{\frac{1}{5}} \\ &= -100 \end{aligned}$$

Question 4

(a) $\frac{1 + \cos \theta}{1 - \cos \theta} = (\csc \theta + \cot \theta)^2$

L.H.S $\frac{1 + \cos \theta}{1 - \cos \theta} \times \frac{(1 + \cos \theta)}{(1 + \cos \theta)}$

$$= \frac{(1 + \cos \theta)^2}{(1 - \cos^2 \theta)^2}$$

$$= \frac{(1 + 2\cos \theta + \cos^2 \theta)}{\sin^2 \theta}$$

$$= \frac{1}{\sin^2 \theta} + \frac{2\cos \theta}{\sin^2 \theta} + \frac{\cos^2 \theta}{\sin^2 \theta}$$

$$= \csc^2 \theta + 2 \frac{\cos \theta}{\sin \theta} \cdot \frac{1}{\sin \theta} + \cot^2 \theta$$

$$= \csc^2 \theta + 2\cot \theta \cdot \csc \theta + \cot^2 \theta$$

$$= (\csc \theta + \cot \theta)^2$$

= RHS

$\therefore \frac{1 + \cos \theta}{1 - \cos \theta} = (\csc \theta + \cot \theta)^2$

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Question 4

$$(b) \quad x = 2 + 3\cos^2\theta$$

$$y = 2\sin\theta - 5$$

$$x = 2 + 3\cos^2\theta$$

$$\frac{x-2}{3} = \cos^2\theta$$

$$\frac{y+5}{2} = \sin\theta$$

$$\frac{(y+5)^2}{4} = \sin^2\theta$$

$$\cos^2\theta + \sin^2\theta = 1$$

$$\frac{x-2}{3} + \frac{(y+5)^2}{4} = 1$$

$$4(x-2) + 3(y+5)^2 = 12$$

$$(c) (i) \sin \frac{x}{2} = \frac{1}{2} \quad 0 \leq x \leq 360^\circ$$

$$0 \leq \frac{x}{2} \leq \frac{360}{2}$$

$$0 \leq \frac{x}{2} \leq 180^\circ$$

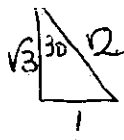
$$\text{let } \theta = \frac{x}{2}$$

$$\therefore \sin \theta = \frac{1}{2} \quad 0 \leq \theta \leq 180^\circ$$

Sin is pos in Q1 + Q2

$$\sin \theta = \frac{1}{2}$$

$$\theta = 30^\circ$$



Quad 1

$$\theta = 30^\circ$$

Quad 2

$$\theta = 180^\circ - 30^\circ$$

$$= 150^\circ$$

$$\therefore \theta = 30^\circ, 150^\circ$$

$$\therefore \frac{x}{2} = 30^\circ, 150^\circ \quad \checkmark$$

$$\therefore x = 60^\circ, 300^\circ \quad \checkmark$$

$$(c) (ii) 2\cos^2\theta - 3\sin\theta - 3 = 0$$

$$\sin^2\theta + \cos^2\theta = 1$$

$$\cos^2\theta = 1 - \sin^2\theta$$

$$2(1 - \sin^2\theta) - 3\sin\theta - 3 = 0$$

$$2 - 2\sin^2\theta - 3\sin\theta - 3 = 0$$

$$-2\sin^2\theta - 3\sin\theta - 1 = 0$$

$$2\sin^2\theta + 3\sin\theta + 1 = 0$$

$$\text{let } x = \sin\theta$$

$$2x^2 + 3x + 1 = 0$$

$$2x^2 + 2x + x + 1 = 0$$

$$2x(x+1) + 1(x+1) = 0$$

$$(2x+1)(x+1) = 0$$

$$x = -\frac{1}{2}, -1$$

$$\therefore \sin\theta = -\frac{1}{2} \text{ or } \sin\theta = -1$$

$$\sin\theta = -\frac{1}{2}$$

Sin neg in Q3 + Q4

$$\sin\theta = -\frac{1}{2}$$

$$\theta = 210^\circ$$

Quad 3

$$\theta = 180^\circ + 30^\circ$$

$$= 210^\circ$$

Quad 4

$$\theta = 360^\circ - 30^\circ$$

$$= 330^\circ$$

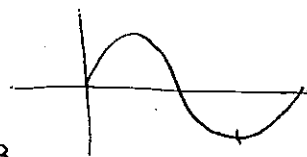
$$\sin\theta = -1$$

$$\sin\theta = 1$$

$$\theta = 90^\circ$$

Sin neg Q2 + Q3

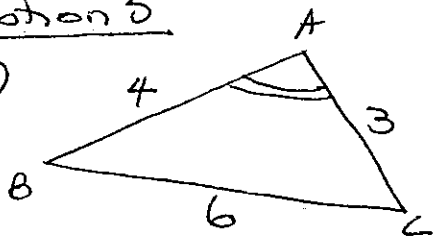
$$\theta = 270^\circ$$



$$\therefore \theta = 210^\circ, 330^\circ, 270^\circ$$

Question 5

(1)



cosine rule $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$

$$\cos A = \frac{4^2 + 3^2 - 6^2}{2 \times 4 \times 3}$$

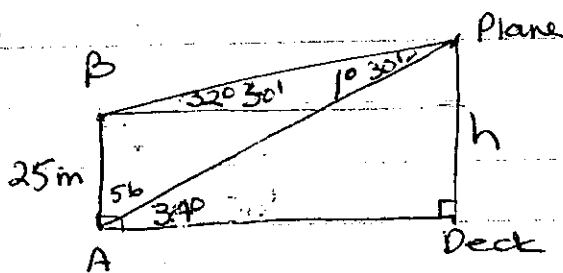
$$\cos A = -0.4583$$

$$A = 117^\circ 17' \text{ (nmin)}$$

$\therefore$ size of the largest angle is $117^\circ 17'$

ii) Area = $\frac{1}{2} ab \sin C$
 $= \frac{1}{2} \times 4 \times 3 \times \sin 117^\circ 17'$
 $= 5.332508665$
 $= 5.3 \text{ cm}^2 \text{ (1dp)}$

(b)



$$\angle BAP = 56^\circ$$

$$\angle ABP = 122^\circ 30'$$

$$\angle BPA = 10^\circ 30'$$

sine rule $\triangle ABP$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$\frac{25}{\sin 10^\circ 30'} = \frac{b}{\sin 122^\circ 30'}$$

$$b = \frac{25 \times \sin 122^\circ 30'}{\sin 10^\circ 30'}$$

$$\therefore AP = 805.4715125$$

In $\triangle APD$

$$\sin 34^\circ = \frac{h}{805.4715125}$$

$$h = 805.4715125 \times \sin 34^\circ$$

$$h = 450.4139577$$

$$\therefore h = 450 \text{ m (n 10.m)}$$

$\therefore$ height of the aircraft is 450m

Question 6

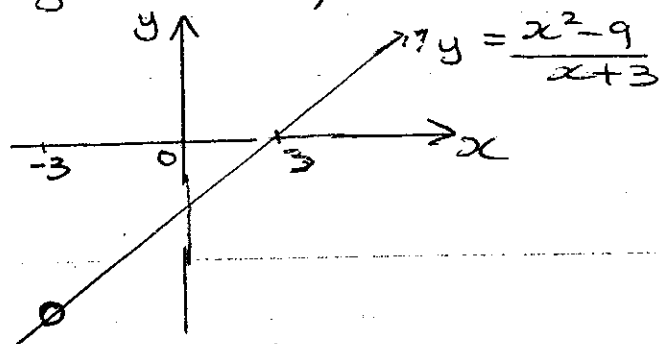
(a) $y = \frac{x^2 - 9}{x + 3} \quad x \neq -3$

$\therefore$ Domain: all real x , $x \neq -3$

$$y = \frac{x^2 - 9}{x + 3}$$

$$y = \frac{(x - 3)(x + 3)}{x + 3}$$

$$y = x - 3, \quad x \neq -3$$



(b) $y = \frac{1 + 2x^2}{1 - x^2}$

Domain: $1 - x^2 \neq 0$
 $(1 - x)(1 + x) \neq 0$
 $x \neq \pm 1$

Vertical asymptotes are $x = 1$, $x = -1$

Horizontal asymptotes:

$$y = \frac{1 + 2x^2}{1 - x^2} \quad (\div x^2)$$

$$y = \frac{\frac{1}{x^2} + \frac{2x^2}{x^2}}{\frac{1}{x^2} - \frac{x^2}{x^2}}$$

$$y = \frac{\frac{1}{x^2} + 2}{\frac{1}{x^2} - 1}$$

$$\lim_{x \rightarrow \infty} \frac{\frac{1}{x^2} + 2}{\frac{1}{x^2} - 1}$$

$$y = -\frac{2}{1}$$

$$y = -2$$

Question 6

b) intercepts

X intercept $y=0$

$$0 = \frac{1+2x^2}{1-x^2}$$

$$0 = 1+2x^2$$

$$-\frac{1}{2} = x^2 \quad \therefore \text{no solutions}$$

$\therefore$ no X intercepts

Y intercept $x=0$

$$y = \frac{1+2(0)^2}{1-(0)^2}$$

$$y = 1$$

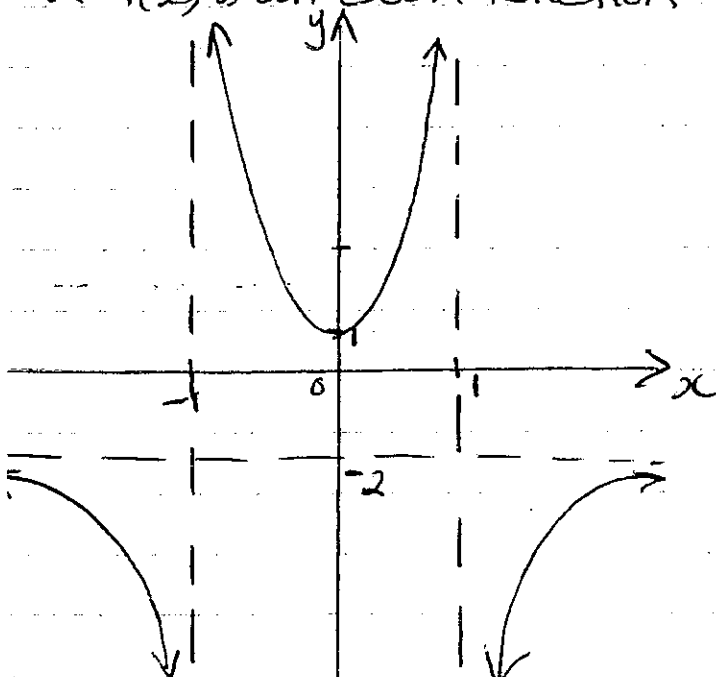
Symmetry

$$f(x) = \frac{1+2x^2}{1-x^2}$$

$$f(-x) = \frac{1+2(-x)^2}{1-(-x)^2}$$

$$= \frac{1+2x^2}{1-x^2}$$

$\therefore f(-x) = f(x)$
 $\therefore f(x)$ is an even function



$$(c) y = f(x) = \frac{1}{\sqrt{x-1}}$$

Domain: $x-1 \geq 0$
 $x \geq 1$

X intercept $y=0$

$$0 = \frac{1}{\sqrt{x-1}} \quad \therefore \text{not possible}$$

$\therefore$ no X intercept

Y intercept $x=0$

$$y = \frac{1}{\sqrt{0-1}} \quad \therefore \text{not possible}$$

$\therefore$ no Y intercept

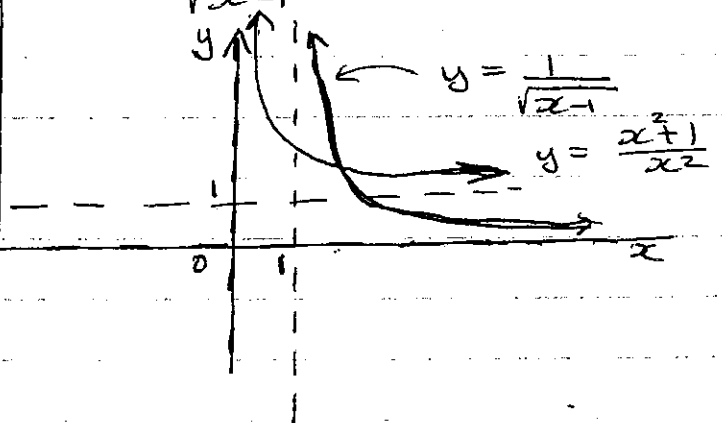
Limit as $x \rightarrow \infty$

$$\lim_{x \rightarrow \infty} \frac{1}{\sqrt{x-1}} \Rightarrow 0$$

Limit as $x \rightarrow 1^+$

$$\sqrt{x-1} \Rightarrow 0$$

$$\frac{1}{\sqrt{x-1}} \Rightarrow \infty$$



(d) Inverse function

$$y = \frac{1}{\sqrt{x-1}}$$

$$x = \frac{1}{y^2-1}$$

$$x^2 = \frac{1}{y^2-1}$$

$$\frac{1}{x^2} = y^2 - 1$$

$$\frac{1}{x^2} + 1 = y$$

$$y = \frac{x^2+1}{x^2}, \quad x \neq 0$$

Domain: $x > 0$
 Range: $y > 1$
 No intercepts